



Artificial neural network prediction of TiO₂-doped chitosan micro/nanoparticle size based on particle imaging measurements

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Abstract

In this study, TiO₂-doped chitosan micro/nanoparticles were fabricated using the ionic gelation mechanism under several process parameters to exhibit the strategy of introducing particle image data for the prediction of particle size. Herein, we report on a detailed methodology for the prediction of prepared particles via artificial neural network (ANN) algorithm using the multi-layer perceptron (MLP) and radial basis function (RBF) models to select the model that demonstrates the best performance for estimation of particle size. Chitosan and TiO₂-doped chitosan micro/nanoparticles were imaged, processed, and analyzed as particle diameters in order to explore prediction models, which were developed under three different classes of prepared particles (chitosan, TiO₂-doped chitosan, and chitosan/TiO₂-doped chitosan). Models were built using particle fabrication process parameters as input with particle size as output. The established MLP model successfully predicted the particle size of all classes with the mean square error (MSE) and correlation coefficient (*R*) between the observed and predicted values in the range of 0.0012–0.0065 and 0.85–0.90, respectively. The best results for prediction were achieved from the RBF model for all classes of particles where MSE and *R* values were determined as 2.93×10^{-22} – 4.93×10^{-11} and 1.0, respectively. Results successfully highlighted the prediction process of particle sizes via MLP and RBF models could be relevant in the decision to produce TiO₂-doped chitosan particles and confirmed the usefulness of particle image data for simulation.

Keywords TiO₂ · Chitosan · Nanoparticles · ANN

Abbreviations

ANN	Artificial neural network
DLS	Dynamic light scattering
FTIR	Fourier transform infrared
MAE	Mean absolute error
MLP	Multi-layer perceptron
MSE	Mean square error
PCL	Poly(caprolactone)
PEG	Polyethylene glycol
PLA	Poly(lactide)
PLGA	Poly(lactic-co-glycolic acid)
PVA	Poly(vinyl acetate)
R	Correlation coefficient
R ²	Coefficient of determination
RBF	Radial basis function
RMSE	Root mean square error
TANSIG	Tangent sigmoid

TiO ₂	Titanium dioxide
TPP	Sodium tripolyphosphate
XRD	X-ray diffraction

List of Symbols

μ_j	RBF function center
σ_j	The spread of the Gaussian basis function
$\phi_j(x)$	The nonlinear function of unit <i>j</i>
x_0	Activation of the hidden layer node
x_n	Activation of <i>n</i> th hidden layer node
$\bar{y}_i$	The mean of the target values
ω_{no}	The interconnection between the <i>n</i> th hidden layer node and the <i>o</i> th output layer node
$f()$	Activation function
$\hat{y}_i$	Network outputs (predicted)
y_i	Targets (observed)

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Introduction

Chitosan is a naturally abundant nontoxic macromolecule polysaccharide that has been reported well by researchers due to its promising properties such as antimicrobial activity

[1–4] against both gram-positive and gram-negative bacteria [5, 6]. On the other hand, titanium dioxide (TiO_2) has been found to be promising, due to its photocatalytic activity, low cost, and biocompatibility, and exhibits antimicrobial activity [1, 7]. The use of TiO_2 and chitosan micro/nanoparticles in several applications including antimicrobial activity is widely advocated. Furthermore, several studies showed that chitosan/ TiO_2 composites exhibited better performance of antibacterial activity than that of the individual components [8–10]. Thus, researches focus on the development of nano/micro scaled particles which can display improvements against resistant bacterium [11], for instance, chitosan- TiO_2 nanocomposites [1, 8] or chitosan-coated TiO_2 nanoparticles [12] with regard to the antibacterial activity.

Controlling size, shape, and morphology is of crucial importance in order to maintain precise fabrication of materials to achieve ideal properties for targeted applications. For example, precisely controlled nanoparticle synthesis is challenging since particle size has a great impact on the success of the application, i.e., drug delivery [13] or tumor targeting [14]. However, the complex synthesis conditions or the contribution of each experimental variable were determined by trial and error based on previous literature knowledge and experience, which makes it difficult to assess the effect of the variables and conditions [15]. Furthermore, the possibility of reaching the universal relations between the efficient recipe variables and resulting particle size necessitates the use of excessive resource, labor, and money [16]. Thus, more efficient and controlled approaches are required for the fabrication of desired nano-sized materials by deriving a mathematical model that can predict the material size or establish necessary processing variables. Developing empirical models such as artificial neural network (ANN) for the prediction of material size would be beneficial since it will decrease time, labor, and material costs [16, 17].

To date, there has been little work focused on investigating the **process parameters** affecting nanoparticle size via ANN models that are explored to be beneficial for skipping many preliminary studies usually required to **optimize formulation** [15, 17–19]. Youshia et al. published a comprehensive study that aimed to build a mathematical model capable of predicting the particle size of polymeric nanoparticles (poly(lactic-co-glycolic acid) (PLGA), poly(vinyl acetate) (PVAc), PCL) [17]. Esmailzadeh-Gharedaghi et al. evaluate and model the effective parameters on the particle size of chitosan nanoparticles using ANNs [15]. Similarly, Hashad et al. used ANNs to identify formulation parameters of chitosan nanoparticles for the prediction of experimental outputs of nanoparticle size and yield [19]. Shalaby et al. conducted ANNs to predict polyethylene glycol/poly(lactide) (PEG/PLA) nanoparticle size and entrapment efficiency of nescapine within the formed nanoparticles

using formulation parameters (molecular weight, ratio of polymer to drug, and number of blocks) [18]. Bozuyuk et al. presented the synthesis of PEGylated chitosan nanoparticles and utilized the ANN model as a tool to predict important properties such as size and zeta potential [20]. However, when the existing studies in the literature are examined, it is clearly seen that the particle size data are obtained from dynamic light scattering (DLS) analysis results. Although DLS is a very powerful tool for the determination of particle sizing since the repeatability of the diameter obtained by DLS is very good, inadequate sample preparation, temperature, and viscosity might lead to a strong distortion of results. The resolution of the DLS technique is limited by the possible agglomeration of particles. Due to these constraints, DLS results are supported with imaging techniques for particle characterization [21].

According to the research hypothesis, the use of an ANN model architecture based on particle image data will enable the prediction of chitosan and TiO_2 -doped chitosan micro/nanoparticle size without the need for excess time, labor, and material costs during particle production which are the main obstacles dedicated to particle manufacturing technology. In this study, we synthesized chitosan and TiO_2 -doped chitosan micro/nanoparticles and prepared particles were imaged and processed by creating a macrocode to measure particle sizes. Then, we introduced the outputs of processed images using ANN algorithms with two developed models (MLP, multi-layer perceptron) and (RBF, radial basis function) for the prediction of particle sizes. The success of MLP and RBF models is compared regarding mean square error (MSE), mean absolute error (MAE), root mean square error (RMSE), correlation coefficient (R), and coefficient of determination (R^2) values; hence, the more efficient and controlled approaches in the production of desired TiO_2 -doped chitosan particle size could be achieved.

Herein, a detailed methodology for utilizing different ANN models is described to allow estimation of the particle size of TiO_2 -doped chitosan micro/nanoparticles. Moreover, this methodology further led to the identification of MLP and RBF models using processed particle image data for particle size prediction methods. This study was not only able to demonstrate the prediction of TiO_2 -doped chitosan micro/nanoparticle sizes via developed models for the first time, but also utilized the strategy of using output data of particle sizes by introducing an imaging system.

Materials and methods

Chitosan (86% deacetylated; medium molecular weight; viscosity 1% in acetic acid 1%, 707 cP; Lot # BCKK4007, Catalogue no. 448877) was purchased from Sigma-Aldrich Chemical Co.

Ltd (Germany). Sodium tripolyphosphate (TPP) (purity 85%, catalogue no. 238503), TiO_2 (titanium (IV) oxide, anatase, nanopowder, <25 nm particle size, 99.7% trace metals basis, catalogue no. 637254), and phosphate-buffered saline (PBS) tablets (pH 7.4) were purchased from Sigma-Aldrich Chemical Co. Ltd (Germany). All other reagents were in analytical grade.

Preparation of chitosan micro/nanoparticles

Chitosan micro/nanoparticles were produced based on ionic gelation of chitosan solutions which are flushed and mixed with TPP solutions under magnetic stirring at room temperature as described before [22]. Briefly, chitosan was dissolved in 1% (v/v) acetic acid solution to maintain chitosan concentrations at 0.25, 0.50, 0.75, 1.0, 1.25, 1.50, 2.0, 2.50, 3.0, 3.5, 4.0, 4.5, and 5.00 (mg/mL). Then chitosan solutions were raised to pH 4.6–4.8 with 1 N NaOH. TPP was dissolved in distilled water to make up TPP solutions of 0.25, 0.50, 0.75, 1.0, 1.25, 1.50, 2.0, 2.50, 3.0, 3.5, 4.0, 4.5, and 5.00 (mg/mL). All solutions were filtered through a 0.22-micron filter (Millipore). The formation of chitosan-TPP micro/nanoparticles started spontaneously via the TPP-initiated ionic gelation mechanism with volumetric ratios of 1:1, 1.5:1, and 2.0:1 (v/v) (chitosan to TPP) under magnetic stirring at room temperature. The formulations of chitosan micro/nanoparticles were conducted at different mixing rates (250, 500, and 750 rpm).

Preparation of TiO_2 -doped chitosan micro/nanoparticles

Chitosan was dissolved in 1% (v/v) acetic acid solution to maintain chitosan concentration at 0.25, 0.50, 0.75, 1.0, 1.25, 1.50, 2.0, 2.50, 3.0, 3.5, 4.0, 4.5, and 5.00 (mg/mL). Prepared chitosan solutions were mixed with TiO_2 at a ratio of TiO_2 to chitosan (1:10) (w/w), and pH was arranged as 4.6–4.8. Prepared TiO_2 -containing chitosan solutions were flushed and mixed with 0.25, 0.50, 0.75, 1.0, 1.25, 1.50, 2.0, 2.50, 3.0, 3.5, 4.0, 4.5, and 5.00 (mg/mL) TPP solutions with a ratio of volume ratio of 1:1, 1.5:1, and 2.0:1 (v/v) (TiO_2 -chitosan to TPP) under magnetic stirring at room temperature. The micro/nanoparticle suspension was gently stirred for 20 min at room temperature to allow excess TiO_2 adsorption on the nanoparticles to reach isothermal equilibrium. The formulations of TiO_2 -doped chitosan micro/nanoparticles were conducted at different mixing rates (250, 500, and 750 rpm).

Imaging micro/nanoparticles and particle diameter size analysis

All prepared particles were viewed under an optical microscope (Leica 295, Germany), and the ImageJ program was utilized for calculating particle diameters. A macro code was created in the ImageJ program to ensure that each

measurement was adjusted due to the scale of the image and then standardized under the same parameters for all images. Images were processed and average particle sizes were determined. Briefly, the images were converted to gray-scale using the “Labstack” to be suitable for the transition to black and white format, the background was removed, and a “Bandpass” filter was applied for resolution. Particle images were processed under black and white scales, and the noise of the image was reduced with the “Median” filter. Then, the sizes of the particles were analyzed by measuring the area of the processed images, and the particle diameter sizes were determined.

Characterization and antibacterial activity of chitosan and TiO_2 -doped chitosan micro/nanoparticles

Chitosan and TiO_2 -doped chitosan micro/nanoparticles having the smallest particle size were used for the analysis of particle characterization and antimicrobial activity. Before analysis, particles were purified by centrifugation at 12,000 g for 30 min. Supernatants were discarded and resuspended in water, and the particles were then freeze-dried. Infrared spectra of chitosan and TiO_2 -doped chitosan micro/nanoparticles were recorded using a Fourier transform infrared spectrophotometer (Bruker IFS 66/S, Germany) in the range of $500\text{--}4000\text{ cm}^{-1}$ to verify the existence of TiO_2 within particles. X-ray diffraction (XRD) patterns of chitosan and TiO_2 -doped chitosan micro/nanoparticles were measured by X-ray diffractometer (PANalytical Empyrean diffractometer), and data were collected over an angular range comprised between 10 and 90° (2θ) with a step size of 0.026° and a dwell time of 93.84 s at each step.

To determine the antimicrobial effect of chitosan and chitosan/ TiO_2 particles, their inhibition areas against gram-positive *Staphylococcus aureus* (American Type Culture Collection No.6538) and gram-negative *Escherichia coli* (American Type Culture Collection No. 25922) were evaluated using the disk diffusion method [23]. Tryptic Soy Broth was used to grow bacterial strains (overnight at 37°C and under aerobic conditions), and then the agar dilution method was assessed with the bacterial strains at 37°C (pH = 7.4) for 24 h. The final bacterial concentration was adjusted to approximately 10^5 CFU/mL . Sterile disks were placed in the middle of Tryptic Soy Agar plates coated with bacterial strains, and $10\text{ }\mu\text{L}$ chitosan and chitosan/ TiO_2 particle solutions were dropped on disks and incubated at 37°C (pH = 7.4) for 24 h. The diameter of the inhibition zone (D, mm) was measured, and all experiments were repeated three times.

Artificial neural network (ANN) modeling

An artificial neural network is a set of interconnected nodes used for understanding and solving modeling

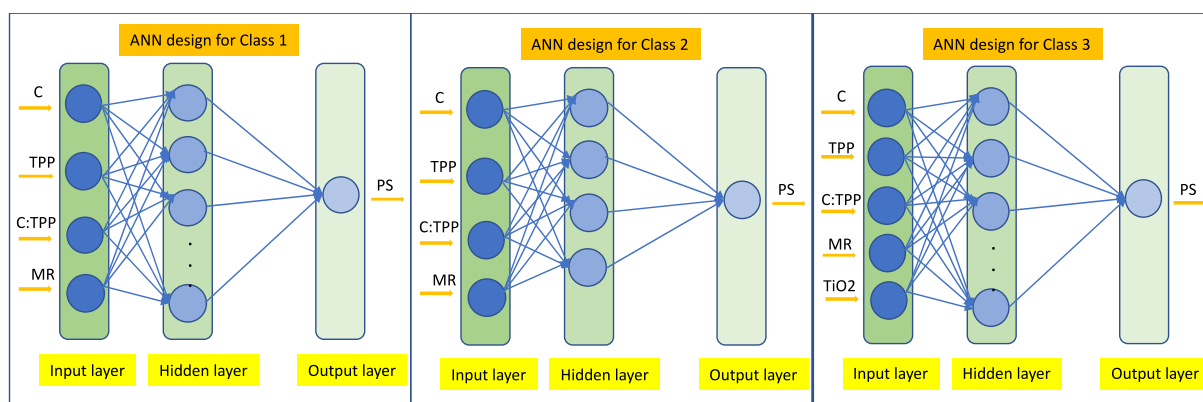


Fig. 1 ANN design architecture for prediction of nanofibers representing Class 1 (chitosan), Class 2 (TiO₂-doped chitosan), and Class 3 (chitosan/TiO₂-doped chitosan). C, chitosan concentration; TPP,

TPP concentration; C to TPP, chitosan to TPP ratio; MR, mixing rate; TiO₂, the presence of TiO₂; PS, the mean particle diameter size

problems that have complex relationships between causal factors and responses [24]. A typical ANN consists of three connected layers: the input layer with independent factors, the output layer with responses, and the hidden layer(s) connecting the input layer to the output one [17]. In this study, 3 classes of input layers (Class1, chitosan particles; Class2, TiO₂-doped chitosan particles; and Class3, chitosan/TiO₂-doped chitosan particles) were evaluated for the responses of particle size. The input layer included 4 independent variables: chitosan concentration, TPP concentration, chitosan to TPP ratio, and mixing rate, for the first and second classes. For the third class where both chitosan and TiO₂-doped chitosan micro/nanoparticles were taken into account, the input layer has 5 independent variables: chitosan concentration, TPP concentration, chitosan to TPP ratio, mixing rate, and the presence of TiO₂. The mean particle size was used as the output layer. For both classes, a hidden layer was fixed for one layer with designated neurons. A web network was designed to connect each neuron to transmit the information between neurons. The prediction of the particle size was performed via designed ANN with the `nntool` command of the MATLAB program (Fig. 1). In this study, the *k*-fold cross-validation method with the `hold-out` command was used to obtain better results.

Data sets, collection, and ANN design

Training and test data sets were created for the attributes of micro/nanoparticles using the data obtained from the ImageJ program. The number of samples to be used in the training and test data sets was randomly divided into parts in the ratio of 80% training to 20% test and applied

to the ANN model. Table 1 shows the ANN architecture and design used for training in our study.

A total of 1521 data were available for the first class of the study (chitosan particles), and 80% of the data (1217 data) was used for the training process and 20% (304 data) for the testing process (data not shown). For the second class (TiO₂-doped chitosan particles) and the third class (total of chitosan and TiO₂-doped chitosan particles), a total of 202 data (162 data for training and 40 data for testing, Supplementary Information) and 1723 data (1379 data for training and 344 data for testing) (data not shown) were used, respectively.

Multi-layer perceptron (MLP) network model, design factors, training parameters

MLP network models are the popular network architectures used in most of the research applications for modeling and prediction [24–27]. Basically, MLP is a feed-forward model comprising one input layer, one or more hidden layers, and one output layer. Each layer has a number of processing units, and each unit is fully interconnected with weighted connections to units in the subsequent layer. The learning algorithm is a process of updating the connection weights of the neurons continuously in hidden layers to minimize the differences between the target and network output samples [28, 29]. Thus, in an MLP neural network with a back-propagation algorithm, the weighted sum of the inputs and bias term is passed to the activation level through a nonlinear transfer function to produce the output which is determined as [26]:

$$x_0 = f\left(\sum_n x_n \omega_{no}\right) \quad (1)$$

Table 1 Simulation parameters

Programme	MATLAB R2016a		
	Class 1	Class 2	Class 3
Number of neurons in input layer	4	4	5
Number of neurons in hidden layer	8	4	8
Number of neurons in output layer	1	1	1
Number of training data sets	1217	162	1379
Number of training data sets	304	40	344
Used ANN model	MLP		
Epoch Number in Training Process	250-500-750-1000		
Neuron number in Training Process	4–40		
Type of Learning algorithm	Levenberg–Marquardt algorithm		
Training Activation function	Hidden Layer (TANSIG)		
	Output Layer (TANSIG)		
Used ANN model	RBF		
Spread Number in Training Process	0.1–1		
Initial Learning Rate	1e–07		
Normalization	Min-Max Normalization		

where $f()$ is the activation function, x_n is the activation of the n^{th} hidden layer node, and ω_{no} is the interconnection between the n^{th} hidden layer node and the output layer node. The most used activation function in the back-propagation algorithm is the sigmoid functions given as [26]:

$$x_0 = \frac{1}{1 + \exp(-\sum x_n \omega_{no})} \quad (2)$$

In this study, the MLP model of neural networks is designed for all classes of particles to perform particle size prediction with the highest accuracy. The data was normalized in the range of [0,1]. The network architecture is assessed by designing the MLP model with the training parameters of a number of neurons, activation function, training function, and number of iterations. The performance of the MLP model was investigated by varying neurons in the range of 4–40. The training function was determined as Levenberg–Marquardt, and the activation function was determined as tangent sigmoid (tansig) for the intermediate layer and output layer. The number of epochs is arranged as 500, 750, and 1000 iterations.

Radial basis function (RBF) network model, design factors, training parameters

RBF neural networks, structured in a similar manner to the MLP model, have a fast learning algorithm, high accuracy, and simple topological structure. However, the radially

symmetric basis function is used as the activation functions of the hidden nodes. The spread constant is basically the Gauss function formula that is characterized by a center and peak width, given as [30]:

$$\phi_j(x) = \exp\left(-\frac{\|x - \mu_j\|^2}{2\sigma_j^2}\right) \quad (3)$$

where $\phi_j(x)$ is the nonlinear function of unit j , μ_j is RBF function center, x the input data vector, and σ_j is the spread of the Gaussian basis function. The spread constant is determined by trial and error in the network structure. The RBF has its peak when the distance between its center and that of the input data vector is zero, and after training, the performance of the network will be defined [26, 27].

In this study, the RBF model of neural networks is designed for all classes of particles to achieve the highest accuracy. The network architecture is developed by designing the RBF model with the training parameters of the number of neurons and the number of spreads. The performance of the RBF model was investigated by varying the spread number in the range of 0–1.0 until the maximum number of neurons is achieved.

Model selection

To evaluate the performance of the designed MLP and RBF models several statistical indicators such as the mean squared error (MSE), the root mean squared error

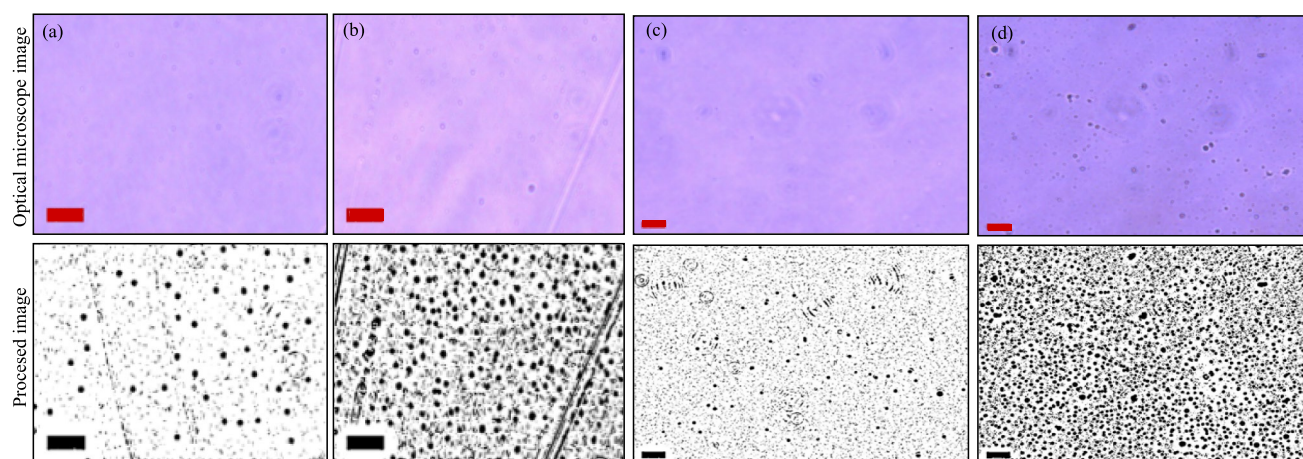


Fig. 2 Optical microscope (40 $\times$) and ImageJ processed images of chitosan micro/nanoparticles prepared under conditions of **a** 0.75 mg/mL chitosan, 5.0 mg/mL TPP, 1.5:1 (v/v) chitosan to TPP ratio, and 250 rpm of mixing rate, scale bar 10 μ m; **b** 1.0 mg/mL chitosan, 4.0 mg/mL TPP, 1:1 (v/v) chitosan to TPP ratio, and 500 rpm of mix-

ing rate, scale bar 10 μ m; **c** 0.5 mg/mL TiO₂-chitosan, 5.0 mg/mL TPP, 1.5:1 (v/v) TiO₂-chitosan to TPP ratio, and 250 rpm of mixing rate, scale bar 10 μ m; **d** 1.25 mg/mL TiO₂-chitosan, 0.25 mg/mL TPP, 1.1:1 (v/v) TiO₂-chitosan to TPP ratio, and 750 rpm of mixing rate, scale bar 10 μ m

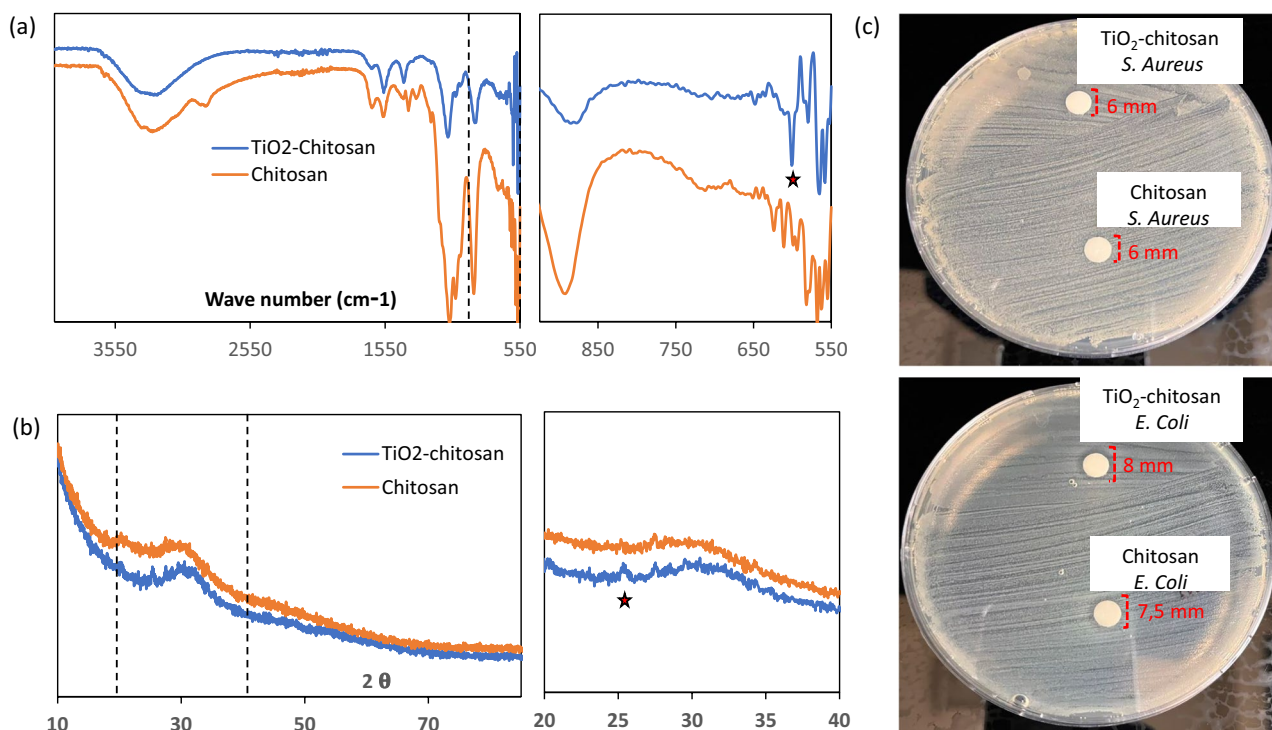


Fig. 3 **a** FTIR spectra and **b** XRD diffraction patterns of chitosan and TiO₂-doped chitosan micro/nanoparticles, **c** inhibition zones of *S. aureus* and *E. coli* indicating the antibacterial activity of chitosan and TiO₂-doped chitosan micro/nanoparticles. Blank disk diameter 6 mm

(RMSE), the mean absolute error (MAE), and the coefficient of determination (R^2) were calculated according to the Eqs. 4–7 [27] presented below:

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n} \quad (4)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (5)$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (6)$$

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (7)$$

where y_i and $\hat{y}_i$ are the targets (observed) and network outputs (predicted) values of the particle diameter size, $\bar{y}_i$ is the mean of the target values, and n represents the number of experiments.

Results and discussion

Chitosan and TiO₂-doped chitosan micro/nanoparticles

The preparation of chitosan particles is based on an ionic gelation interaction between positively charged chitosan and negatively charged TPP [31]. Particle size and size distribution of the chitosan depend largely on the concentration of chitosan and TPP solutions, chitosan to TPP (v/v) ratio, and mixing [22, 32–34]. Chitosan micro/nanoparticles were successfully prepared via an ionic gelation mechanism using four different

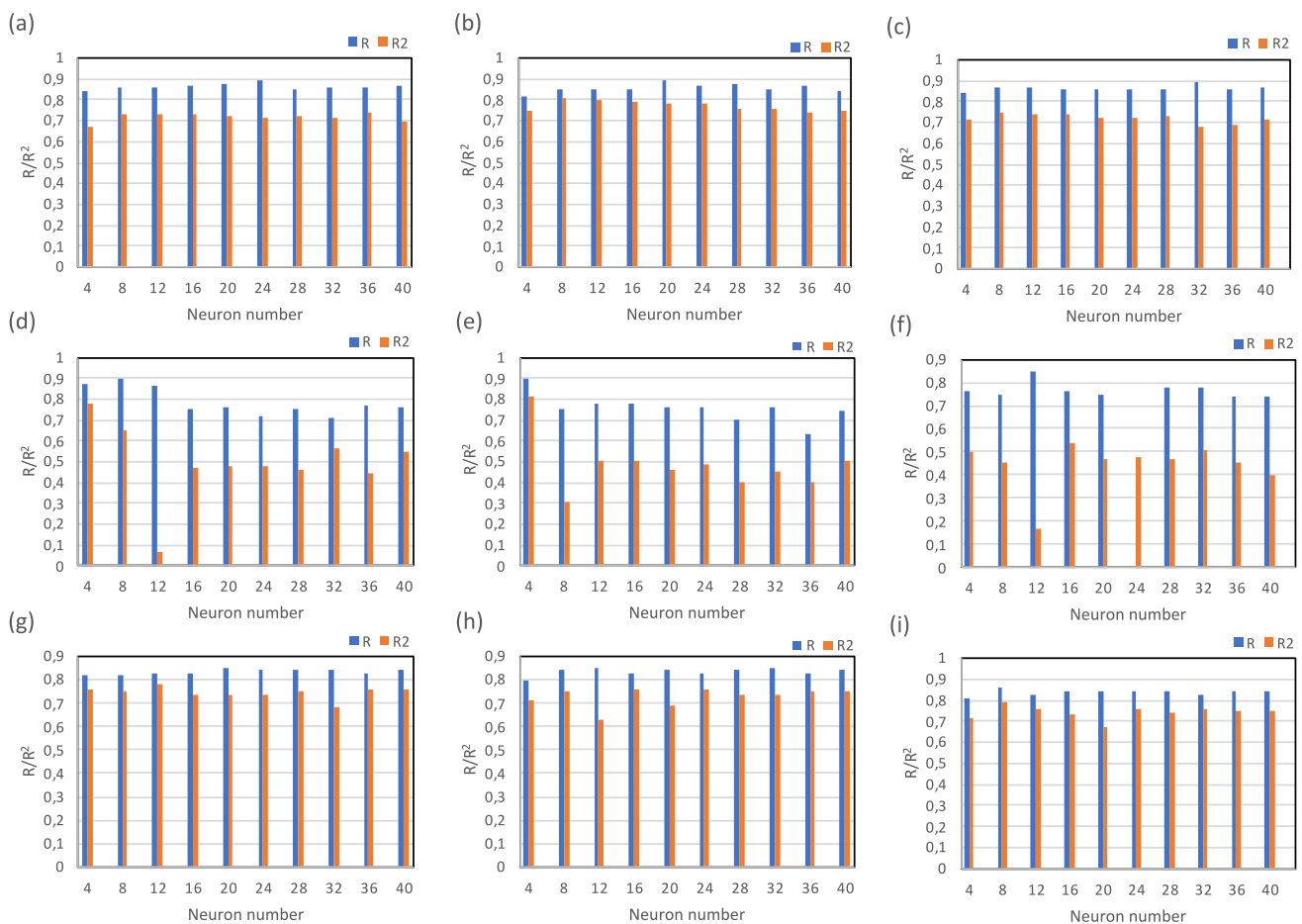
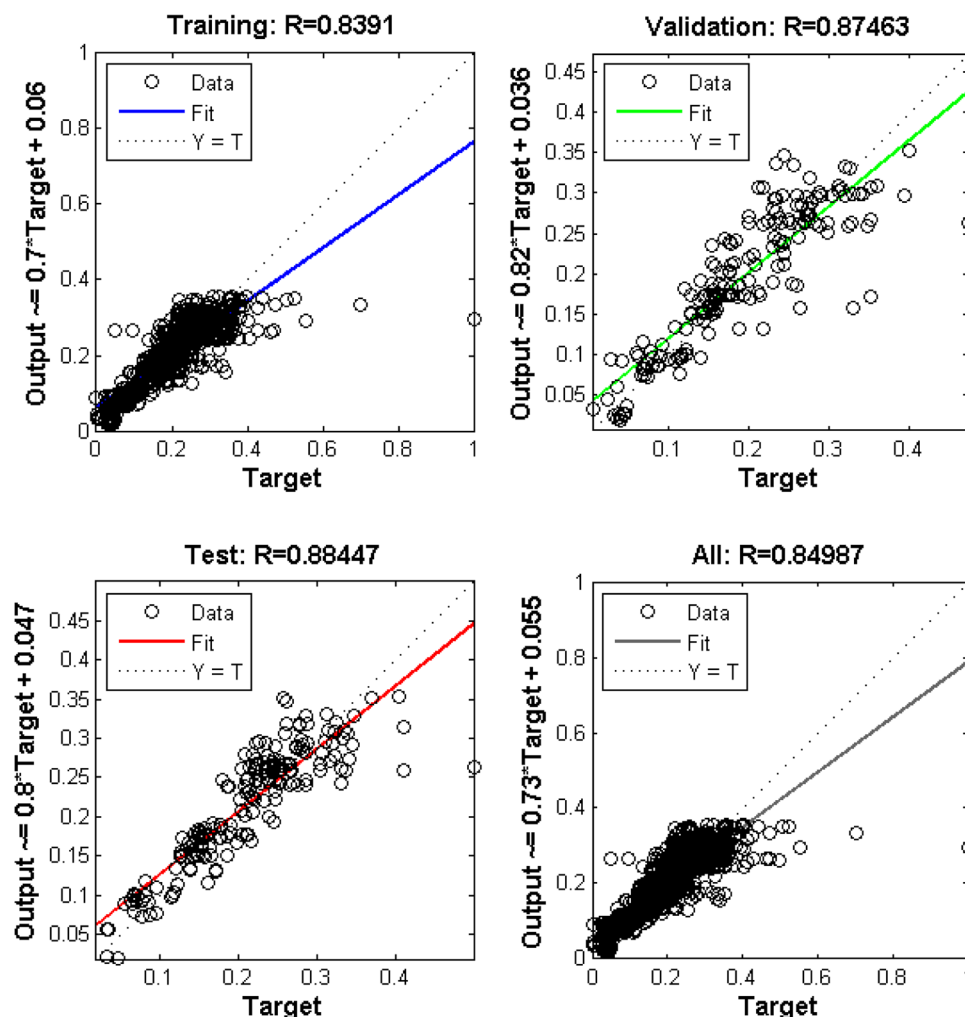


Fig. 4 The results of the designed MLP model for the size of chitosan micro/nanoparticle prediction; **a–c** with epoch number of **a** 500, **b** 750, and **c** 1000, TiO₂-doped chitosan micro/nanoparticle prediction;

d–f with epoch number of **d** 250, **e** 500, and **f** 750, chitosan/TiO₂-doped chitosan micro/nanoparticle prediction; **g–i** with epoch number of **g** 250, **h** 500, and **i** 750

Fig. 5 Plots of output versus target predict the size of chitosan micro/nanoparticle values by MLP network model



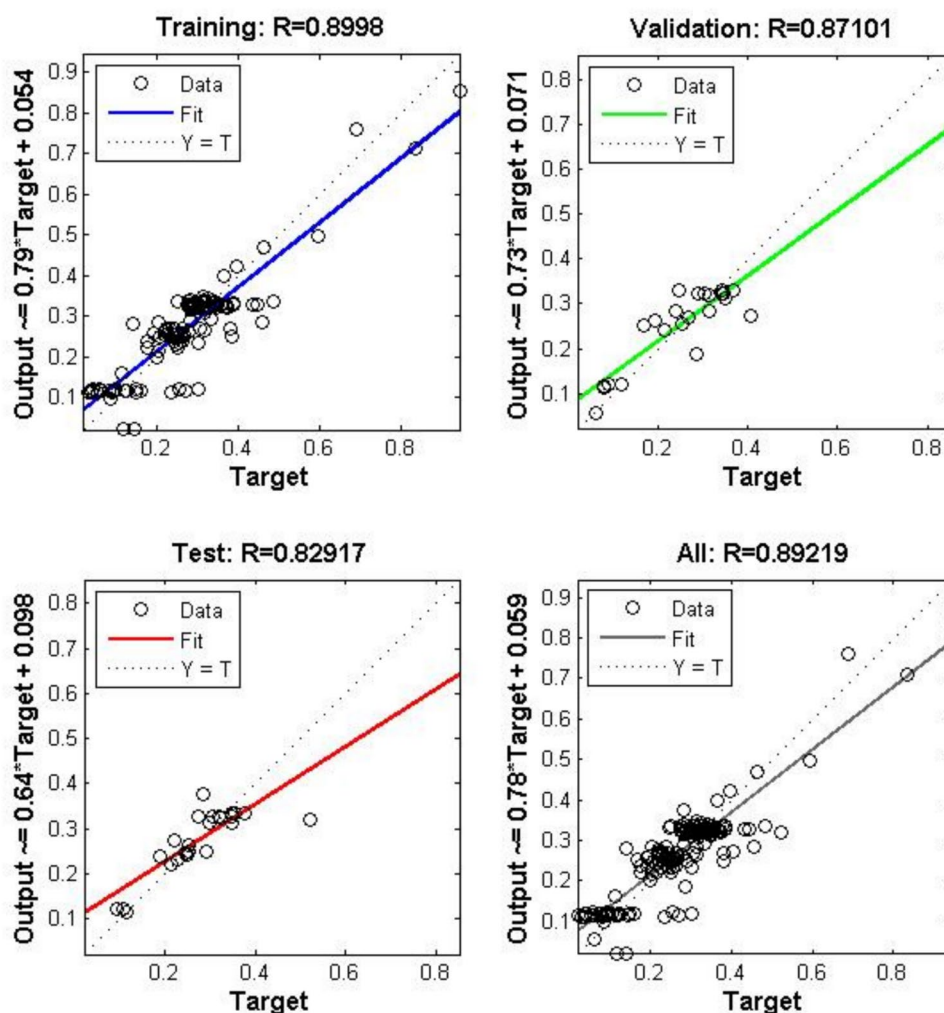
particle fabrication parameters. The formulation was assessed by varying different numbers of parameters of chitosan concentration, TPP concentration, chitosan to TPP ratio, and mixing rate (13, 13, 3, and 3, respectively). A total of 1521 data sets of chitosan micro/nanoparticles were maintained that were available for particle imaging and processing.

Figure 2a, b shows optical images and ImageJ processed images of chitosan micro/nanoparticles having the smallest average particle diameter (322.3 nm) (Fig. 2a) and the highest average particle diameter (1217.4 nm) (Fig. 2b) fabricated under the designated formulations. Figure 2a represents the chitosan micro/nanoparticles prepared under the formulation parameters of 0.75 mg/mL chitosan, 5.0 mg/mL TPP, 1.5:1 (v/v) chitosan to TPP ratio, and 250 rpm of mixing rate. On the other hand, the highest average particle diameter of chitosan particles was fabricated using the formulation parameters of 1.0 mg/mL chitosan, 4.0 mg/mL TPP, 1:1 (v/v) chitosan to TPP ratio, and 500 rpm of mixing rate (Fig. 2b).

TiO₂-doped chitosan micro/nanoparticles were fabricated via the same method as in chitosan particles. The

formulation parameters of TiO₂-chitosan concentration, TPP concentration, TiO₂-chitosan to TPP ratio, and mixing rate are readily used similar to the chitosan particle formulation, chosen with regard to the smallest average particle diameter of chitosan particles. Thus, a total of 202 data sets of TiO₂-doped chitosan micro/nanoparticles were maintained that were available for particle imaging and processing. Figure 2c, d demonstrates the optical microscope and ImageJ processed images of TiO₂-doped chitosan micro/nanoparticles. The smallest average particle diameter of TiO₂-doped chitosan micro/nanoparticles was determined (427.3 nm) from the processed image obtained from the particles prepared under the formulation parameters of 0.5 mg/mL TiO₂-chitosan, 5.0 mg/mL TPP, 1.5:1 (v/v) TiO₂-chitosan to TPP ratio, and 250 rpm of mixing rate (Fig. 2a). Figure 2b represents the highest average particle diameter of TiO₂-doped chitosan micro/nanoparticles (1137.2 nm) fabricated using the formulation parameters of 1.25 mg/mL TiO₂-chitosan, 0.25 mg/mL TPP, 1.1:1 (v/v) TiO₂-chitosan to TPP ratio, and 750 rpm of mixing rate.

Fig. 6 Plots of output versus target predict the size of TiO₂-doped chitosan micro/nanoparticle values by MLP network model



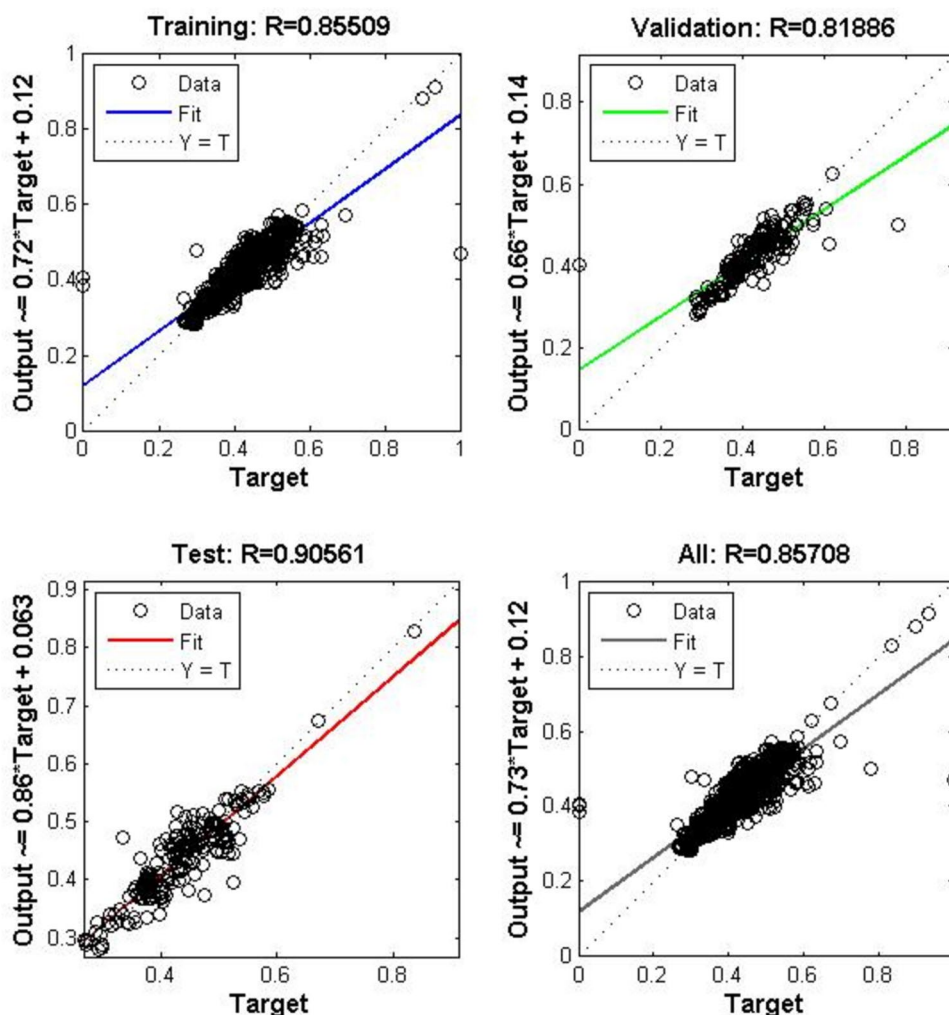
Characterization of chitosan and TiO₂-doped chitosan micro/nanoparticles

Structural characterization of chitosan and TiO₂-doped chitosan micro/nanoparticles was evaluated by Fourier transform infrared (FTIR) studies. FTIR spectra of chitosan and TiO₂-doped chitosan micro/nanoparticles are presented in Fig. 3a. The presence of TiO₂ in TiO₂-doped chitosan micro/nanoparticles was confirmed with a peak at $\sim 600 \text{ cm}^{-1}$ due to the Ti–O stretching vibration indicating the presence of TiO₂ nanoparticles that interact with the chitosan matrix [35, 36].

The crystalline structure of particles was examined by X-ray diffraction (XRD) technique, and XRD diffraction patterns of chitosan and TiO₂-doped chitosan micro/nanoparticles are shown in Fig. 3b. As seen in XRD patterns, the diffraction peak at 2θ of 25.44° was observed for TiO₂-doped chitosan micro/nanoparticles (Fig. 3b) that confirms TiO₂ in chitosan particles had the crystal structure assigned to anatase phase titanium (JCPDS 21–1272) [37, 38].

Chitosan and TiO₂-chitosan composites have been investigated for their antibacterial activity that was widely determined by gram-positive and gram-negative bacteria [9, 39, 40]. However, chitosan has been proven to be more effective against gram-negative bacteria than gram-positive bacteria [9]. Moreover, the inhibitory activity of TiO₂ nanoparticles for *S. aureus* and *E. coli* was reported with the best selectivity against gram-negative (*E. coli*) bacteria [41]. In this study, the zone inhibition test was performed to determine the inhibition area for *S. aureus* and *E. coli*, represented in Fig. 3c. Results showed that no inhibition zone is detected for chitosan and chitosan/TiO₂ particles against *S. Aureus* bacterial type. However, a zone formation against *E. Coli* was detected from 6 mm disc diameter (blank disc) to 7.5 mm and 8 mm disc diameter in the presence of chitosan and chitosan/TiO₂ particles, respectively (Fig. 3c). Results indicate the antimicrobial activity against *E. coli*, suggesting the formulated usage of chitosan/TiO₂ particles.

Fig. 7 Plots of output versus target predict the size of chitosan/TiO₂-doped chitosan micro/nanoparticles values by MLP network model



MLP network modelling

In the learning algorithm, the connection weights of the neurons are updated continuously in hidden layers where each layer has a number of processing units. Previously, our preliminary results showed that the MLP model, designed with varying neurons in hidden layers, has good predictive capabilities for the prediction of chitosan particle size (data not shown). Thus, in this study, the MLP model of neural networks is designed for all classes of particles given in Table 1 to perform particle size prediction with the highest accuracy. Coefficient of determination (R^2) and correlation coefficient (R) values were determined according to the epoch numbers with varying neurons in hidden layers.

Prediction performance of chitosan micro/nanoparticles

Figure 4a–c demonstrates R^2 and R values obtained from MLP modelling of chitosan micro/nanoparticles. Results show that the highest R^2 and R values were achieved for 750 iterations of processing, and the maximum R^2 value (0.81)

was determined to be 8 neurons in the hidden layer. Moreover, the variations in R^2 values occurred with the change in the number of neurons in each processing (Fig. 4a–c).

The regression performance graphs for chitosan micro/nanoparticle size prediction are presented in Fig. 5. The scatter plots provide the correlation between variables which is used to define the accuracy of the MLP model. The regression results from MLP outputs versus the target values of the chitosan micro/nanoparticles for the training, validation, testing, and overall data were calculated as 0.8391, 0.87463, 0.88447, and 0.84987, respectively (Fig. 5). According to the best prediction performance results ($R^2 = 0.8091$), the error analysis was performed, and the calculated MSE, MAE, and RMSE values are 0.0016, 0.0306, and 0.0402, respectively.

Prediction performance of TiO₂-doped chitosan micro/nanoparticles

R^2 and R values were obtained from MLP modelling of TiO₂-doped chitosan micro/nanoparticles and were determined

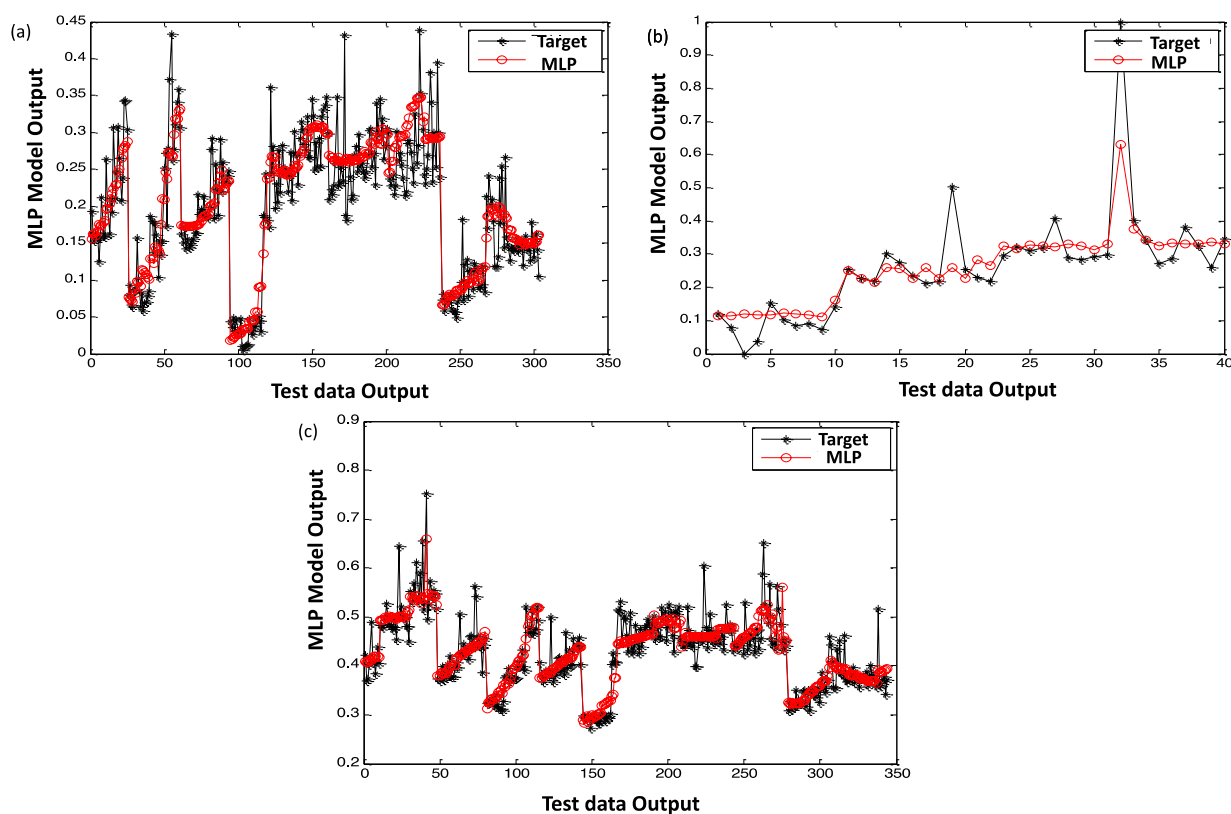


Fig. 8 Target and simulated size of **a** chitosan, **b** TiO_2 -doped chitosan, **c** chitosan/ TiO_2 -doped chitosan micro/nanoparticle test data by MLP network model

according to the epoch numbers with varying neurons in hidden layers (Fig. 4d–f). Results show that the best R^2 and R values were achieved for 500 iterations of processing, and the maximum R^2 value (0.81) was determined to be 4 neurons in the hidden layer. Figure 4d–f shows that R^2 values dramatically changed with the number of neurons in each processing.

TiO_2 -doped chitosan micro/nanoparticle size prediction performance is demonstrated by the regression performance graphs presented in Fig. 6.

The scatter plots show MLP outputs versus the target values of the TiO_2 -doped chitosan micro/nanoparticles. According to the results from MLP outputs versus the target values, the training, validation, testing, and overall regression data were calculated as 0.8998, 0.87101, 0.82917, and 0.89219, respectively (Fig. 6). According to the best prediction performance results ($R^2=0.8104$), the error analysis was performed, and the calculated MSE, MAE, and RMSE values are 0.0065, 0.0462, and 0.0809, respectively.

Prediction performance of Chitosan/ TiO_2 -doped chitosan micro/nanoparticles

Figure 4g–i shows R^2 and R values obtained from MLP modelling for the prediction of particle sizes of all particles

namely in the class of “Chitosan/ TiO_2 -doped chitosan micro/nanoparticles.” According to the results, the best R^2 and R values were achieved for 750 iterations of processing, and the maximum R^2 value (0.79) was determined to be 8 neurons in the hidden layer (Fig. 4g–i). A similar change with varying neurons in hidden layers was determined in R^2 values (Fig. 4g–i) compared to R^2 values (Fig. 4a–c) obtained from the class of chitosan particles.

Figure 7 shows the scatter plots of MLP outputs versus the target values of the chitosan/ TiO_2 -doped chitosan micro/nanoparticles indicating the particle size prediction performance. The training, validation, testing, and overall regression data were calculated as 0.85509, 0.81886, 0.90561, and 0.85708, respectively (Fig. 7). According to the best prediction performance results ($R^2=0.7845$), MSE, MAE, and RMSE values are calculated as 0.0012, 0.0239, and 0.0347, respectively.

Validation performance of MLP model outputs

In order to confirm the compatibility of the test data with the calculations in the MLP model, the validation performance of the target and predicted outputs of all classes of particles are demonstrated in Fig. 8. The presented graph (Fig. 8a) shows a meaningful and distinctive agreement

Table 2 The value of the error parameters for tuning in the RBF network model

Spread number	MAE	MSE	RMSE	R^2	R
Chitosan micro/nanoparticles					
0.1	4.70E-12	2.93E-22	1.71E-04	1	1
0.2	7.22E-04	5.67E-06	2.38E-03	0.99934	0.99967
0.3	0.00853	0.00026	0.01611	0.96964	0.98470
0.4	0.01383	0.00061	0.02461	0.92915	0.96393
0.5	0.01698	0.00081	0.02845	0.90538	0.95151
0.6	0.01961	0.00101	0.03180	0.88176	0.93902
0.7	0.02109	0.00116	0.03409	0.86413	0.92959
0.8	0.02207	0.00124	0.03519	0.85515	0.92475
0.9	0.02300	0.00133	0.03640	0.84504	0.91936
1.0	0.02396	0.00140	0.03736	0.83680	0.91477
TiO ₂ -doped chitosan micro/nanoparticles					
0.1	4.72E-13	4.93E-11	7.02E-13	1	1
0.2	6.98E-06	7.75E-11	8.80E-06	1	1
0.3	0.00582	1.63E-04	0.01275	0.99116	0.99557
0.4	0.01197	5.89E-04	0.02428	0.96795	0.98385
0.5	0.01834	0.00104	0.03225	0.94347	0.97132
0.6	0.02180	0.00122	0.03490	0.93377	0.96666
0.7	0.03026	0.00235	0.04849	0.87215	0.93393
0.8	0.03316	0.00304	0.05515	0.83462	0.91358
0.9	0.03557	0.00319	0.05651	0.82637	0.90924
1.0	0.03736	0.00353	0.05937	0.80836	0.89909
Chitosan/TiO ₂ -doped chitosan micro/nanoparticles					
0.1	6.60E-12	6.55E-22	2.56E-11	1	1
0.2	9.77E-04	2.05E-05	4.53E-03	0.99666	0.99833
0.3	0.00710	1.97E-04	0.01405	0.96787	0.98380
0.4	0.01118	4.31E-04	0.02077	0.92978	0.96425
0.5	0.01363	5.75E-04	0.02399	0.90632	0.95201
0.6	0.01573	7.34E-04	0.02708	0.88057	0.93840
0.7	0.01729	8.70E-04	0.02949	0.85839	0.92649
0.8	0.01787	9.21E-04	0.03035	0.85000	0.92196
0.9	0.01892	9.86E-04	0.03140	0.83944	0.91621
1.0	0.01944	0.00106	0.03250	0.82806	0.90998

Bold entries indicate highest R values

between the target (observed) and simulated (predicted) values of the chitosan micro/nanoparticles with the MLP network model. Similarly, a notable and satisfying agreement between the target (observed) and simulated (predicted) values of the TiO₂-chitosan (Fig. 8b) and chitosan/TiO₂-chitosan (Fig. 8c) micro/nanoparticles indicated the validation performance of MLP network model.

RBF network modelling

RBF model of neural networks is designed for all classes of particles to achieve the highest accuracy with varying

neurons with the spread constants (0.1–1), determined by trial and error in the network structure. The best R^2 and R values were achieved as the maximum number of neurons converges to a total number of data used for all classes of particles. Thus, the RBF model was designed with 1521 neurons, 202 neurons, and 1723 neurons for the prediction of particle sizes of chitosan, TiO₂-doped chitosan, and chitosan/TiO₂-doped chitosan particles, respectively.

Prediction performance of chitosan, TiO₂-doped chitosan, and chitosan/TiO₂-doped chitosan particles

Table 2 presents the effect of the number of spreads on RBF modelling, which was designed with their maximum number of neurons for the prediction of particle sizes of chitosan, TiO₂-doped chitosan, and chitosan/TiO₂-doped chitosan particles. Results indicated that as the number of spread increased, the error values were dramatically increased. Moreover, the best R and R^2 values were achieved for the smallest spread numbers for all classes of particles at their maximum number of neurons.

Validation performance of RBF model outputs

The compatibility of the test data with the calculations in the RBF model is confirmed by the validation performance of the target and predicted outputs demonstrated in Fig. 9. The extremely consistent target (observed) and simulated (predicted) values were determined for all classes of particles (Fig. 9a–c). Thus, the performance of the test data with regard to the simulations in the RBF model indicates the validation of the RBF model with artificial neural networks.

ANN model comparison

ANNs consist of a large class of different architectures, and MLP and RBF are two of the most widely used in literature for classification or regression problems with the ability to generalize for imprecise input data. MLP and RBF models are both good in prediction studies; however, RBF is a localist type of learning which is responsive only to a limited section of input space. On the other hand, MLP is a more distributed approach [26, 42]. Previously, Yilmaz et al. reported that the constructed RBF exhibited a higher performance than MLP for predicting the swell percent of soil in their study [26]. In this study, MLP and RBF models were used and compared for the prediction of particle sizes obtained from particle imaging. The comparison of the RBF and MLP network models indicates the good predictive capabilities of the RBF model with higher accuracy and extremely lower error values than the MLP model (Table 3).

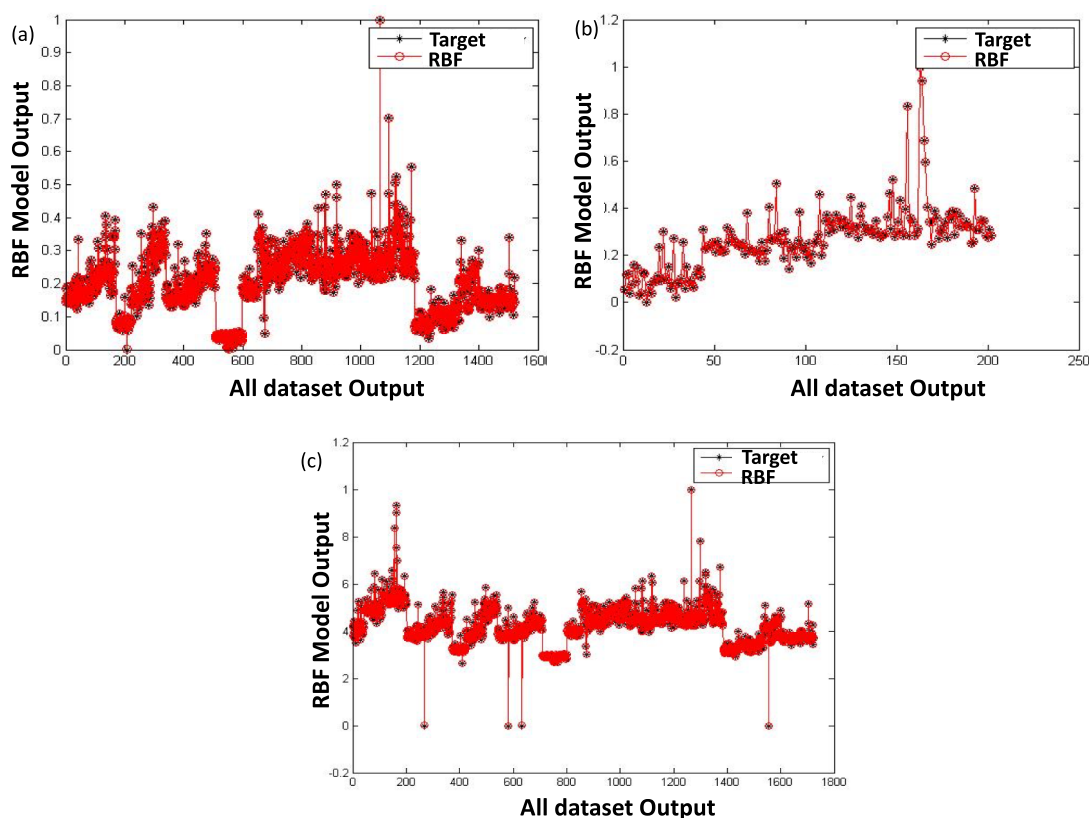


Fig. 9 Target and simulated size of **a** chitosan, **b** TiO₂-doped chitosan, **c** chitosan/TiO₂-doped chitosan micro/nanoparticle all dataset by RBF network model

Conclusion

In particulate formulations of chitosan and their composites via ionic gelation mechanism, the fabrication parameters (chitosan concentration, TPP concentration, Chitosan to TPP ratio, and mixing rate) are important factors to determine the required particle size for several applications. Therefore, using accurate fabrication parameters by simulation through appropriate models is desired and is of vital importance since the determination of required particle size is time-consuming and requires a lot of labor work and specialized facilities in the laboratory. In literature, several computing models were studied and evaluated the performances for predicting the particle size of chitosan. Hashad et al. measured chitosan particle size by DLS technique, the predicted particle size, and error values (R^2 and RMSE values of 0.9835 and 14.54, 0.9658 and 6.55, and 0.952 and 19.13 for training, test, and validating data, respectively) calculated by the MLP model [19]. Esmailzadeh-Gharedaghi et al. also used DLS data of chitosan particles and modeled the data by ANNs for size prediction, resulting in the best R^2 values of 0.87, 0.98, and

0.82 for the training, test, and validation data, respectively [15]. Baharifar et al. used DLS data of albumin-loaded chitosan nanoparticles and reported the performance of the ANN model (R^2 of the train, test, and unseen data was 0.97, 0.95, and 0.88, respectively) for predicting the relationship of parameters [43]. Kizilel et al. studied size prediction of PEGylated chitosan nanoparticles using DLS

Table 3 Best performance indices (MSE, MAE, RMSE, and R^2) for models

	MSE	MAE	RMSE	R^2
MLP model				
Chitosan	0.0016	0.0306	0.0402	0.8091
TiO ₂ -doped chitosan	0.0065	0.0462	0.0809	0.8104
Chitosan/TiO ₂ -doped chitosan	0.0012	0.0239	0.0347	0.7845
RBF model				
Chitosan	2.93E-22	4.70E-12	1.71E-04	1
TiO ₂ -doped chitosan	4.93E-11	4.72E-13	7.02E-13	1
Chitosan/TiO ₂ -doped chitosan	6.55E-22	6.60E-12	2.56E-11	1

data with MLP model and reported R^2 values as 0.8734 and 0.7869 for train and test data, respectively [20].

In this study, we used particle image data of chitosan and TiO₂-doped chitosan particles for measuring particle sizes using the ImageJ program with a macro code for calculations. The performance of MLP and RBF models for particle size prediction was evaluated for 3 classes of particles studied. According to the performance metrics, successful prediction of particle sizes via particle image data was achieved for all classes using the proposed models (R^2 0.78–1.0). The research notably introduces a powerful approach for the prediction of chitosan and TiO₂-doped chitosan micro/nanoparticle size with the ANN strategy by using particle image data. Moreover, the study highlights the comparison of the RBF and MLP network models which indicates the good predictive capabilities of the RBF model. Thus, the practical outcome of the proposed models could be used, with acceptable accuracy, at the preliminary stage of the design of chitosan and TiO₂-doped chitosan micro/nanoparticles.

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Data availability The TiO₂-doped chitosan particle data supporting this article have been included as part of the Supplementary Information, and all other data are available on request from the authors.

Declarations

Ethical approval Not applicable.

Competing interests The authors declare no competing interests.

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